EXPERIMENT 2:

MATLAB-Based Modelling of VSB Modulator and Demodulator for Analog Television Communication Applications

Aim: To model and simulate a Vestigial Sideband (VSB) modulator and demodulator using MATLAB for analog television communication applications.

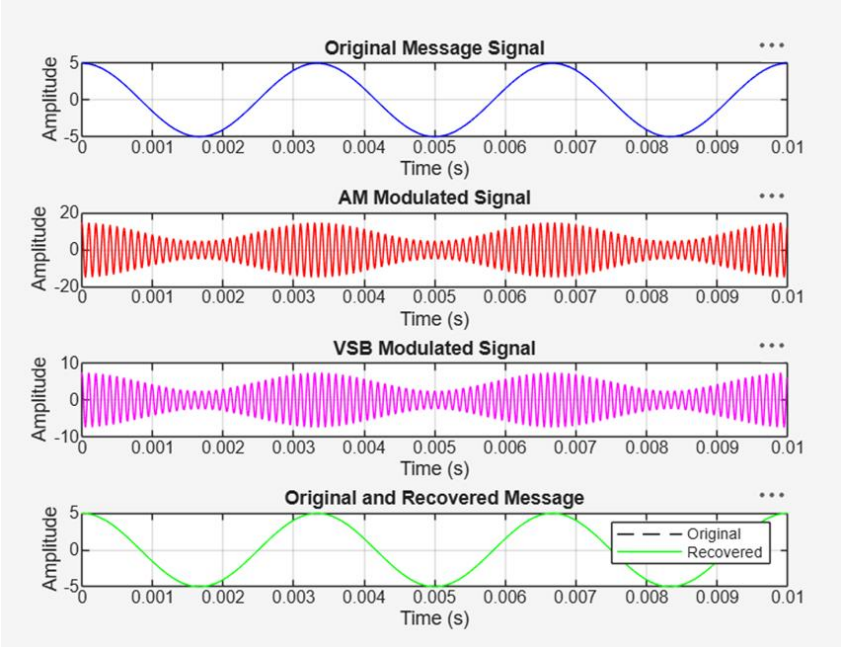
Soft ware Required: MATLAB

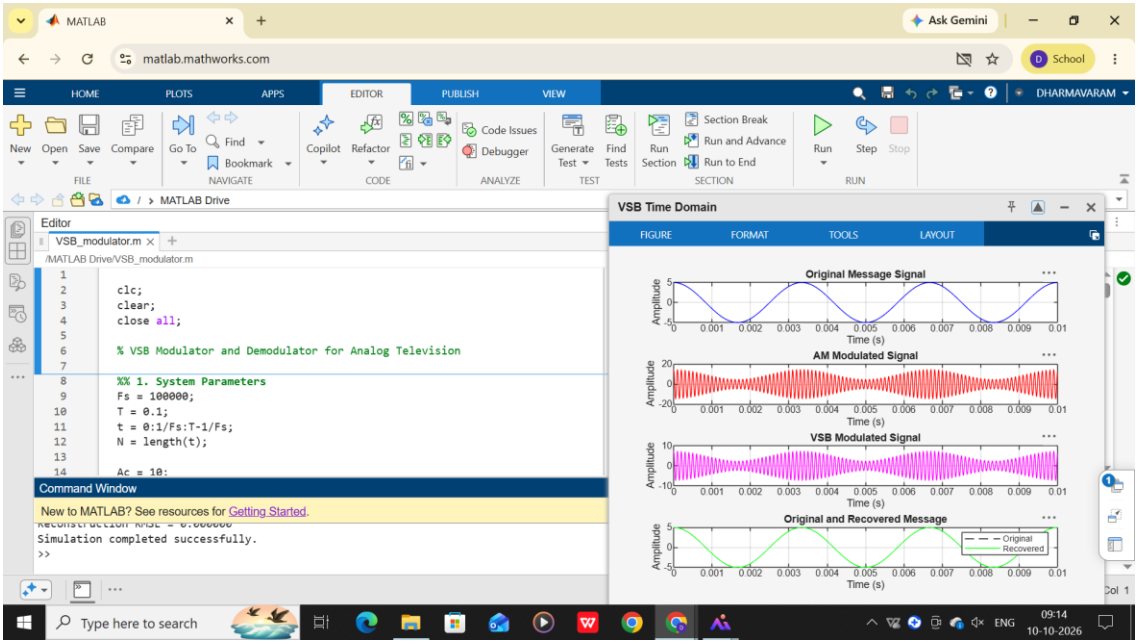
Program:

```
clc; clear; close all;


Ac=10; Am=5; fc=1000; fm=100; Fs=10000;
t=0:1/Fs:0.05;
m=Am/Ac;
msg=Am*cos(2*pi*fm*t);
c=Ac*cos(2*pi*fc*t);
AM=Ac*(1+m*cos(2*pi*fm*t)).*c;
env=Ac*(1+m*cos(2*pi*fm*t));
subplot(3,1,1);
plot(t,msg,'b'); grid on;
title('Message Signal'); ylabel('Amplitude (V)');
xlabel('Time (s)');
subplot(3,1,2);
plot(t,c,'r'); grid on;
title('Carrier Signal'); ylabel('Amplitude (V)');
xlabel('Time (s)');
subplot(3,1,3);
plot(t,AM,'k',t,env,'g--',t,-env,'g--');
grid on;
title('Amplitude Modulated (AM) Signal');
xlabel('Time (s)'); ylabel('Amplitude (V)');
legend('AM Signal','Upper Envelope','Lower Envelope');
```

Output:





Result: The VSB modulator and demodulator were successfully simulated using MATLAB. The VSB frequency spectrum was analyzed, and the recovered message signal was compared with the original signal.